Final Exam Practice — Linear Algebra

COMP3670/6670

Show your calculations and justify your answers. Questions marked (*) require a proof. All vectors and matrices are real. Vector norms are Euclidean; $\|\cdot\|_F$ denotes the Frobenius norm. Use the covariance convention $\mathbf{C} = \frac{1}{n}\mathbf{X}_c^\mathsf{T}\mathbf{X}_c$, where the transposes of the centered datapoints are the rows of $\mathbf{X}_c$.

1. Classify each matrix below as (i) not in row-echelon form, (ii) in row-echelon form but not reduced row-echelon form or (iii) in reduced row-echelon form. Give a reason for each classification. For each matrix not in class (iii), perform row operations to obtain its reduced row-echelon form.

$$\mathbf{A} = \begin{pmatrix} 2 & 4 & 1 & 3 & 3 \\ 0 & 0 & -1 & 2 & 1 \\ 0 & 0 & 0 & 3 & 6 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 3 & 0 & -2 \\ 0 & 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\mathbf{D} = \begin{pmatrix} 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \mathbf{E} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \mathbf{F} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 2 & 0 \end{pmatrix},$$

$$\mathbf{G} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \mathbf{H} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

2. Consider the linear system, with parameters $a, b \in \mathbb{R}$,

$$\begin{aligned} x_1 + x_2 + x_3 &= 2, \\ 2x_1 + 3x_2 + 4x_3 &= 5, \\ x_1 + 2x_2 + (a+2)x_3 &= b. \end{aligned}$$

Use Gaussian elimination to determine exactly when the system has no solution, a unique solution or infinitely many solutions. Give the solution set in every consistent case.

3. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 1 & 2 & 1 & 3 \end{pmatrix}.$$

(a) Find the reduced row-echelon form of $\mathbf{A}$ and identify its pivot columns.

(b) Find bases of $\mathrm{Col}(\mathbf{A})$ and $\mathrm{Nul}(\mathbf{A})$. State the ambient space and dimension of each.

(c) Give a condition on $\mathbf{b} = (b_1, b_2, b_3)^\mathsf{T}$ for $\mathbf{Ax} = \mathbf{b}$ to be consistent. Can a consistent system with this coefficient matrix have a unique solution?

4. A matrix $\mathbf{M} \in \mathbb{R}^{4\times 7}$ has a null space of dimension 3.

(a) Find its rank and describe its column space.

(b) For each $\mathbf{b} \in \mathbb{R}^4$, does $\mathbf{Mx} = \mathbf{b}$ have no solution, a unique solution or infinitely many solutions? Explain.

(c) Given a solution $\mathbf{x}_0$ of $\mathbf{Mx} = \mathbf{b}$, express all solutions using $\mathbf{x}_0$ and a basis $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ of $\mathrm{Nul}(\mathbf{M})$.

5. Let

$$\mathbf{S} = \begin{pmatrix} 3 & 2 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 2 & 3 \end{pmatrix}, \qquad \mathbf{v}(s,t) = \begin{pmatrix} s \\ 1 \\ t \\ 1 \end{pmatrix}.$$

(a) Find all pairs $(s,t) \in \mathbb{R}^2$ for which $\mathbf{v}(s,t)$ is an eigenvector of $\mathbf{S}$. Find the eigenvalue for each such pair.

(b) Check that the two eigenvectors you found are orthogonal.

(c) Find the eigenvalues corresponding to $(2,1,-2,-4)^{\mathsf{T}}$ and $(-2,4,2,-1)^{\mathsf{T}}$, and write an orthonormal eigenbasis of $\mathbf{S}$.

6. Let

$$\mathbf{q}_1 = \frac{1}{3}\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}, \qquad \mathbf{q}_2 = \frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}.$$

(a) Verify that $\mathbf{q}_1$ and $\mathbf{q}_2$ are orthonormal.

(b) Find all vectors $\mathbf{q}_3$ that complete them to an orthonormal basis of $\mathbb{R}^3$.

(c) Choose one such $\mathbf{q}_3$, form $\mathbf{Q} = (\mathbf{q}_1 \ \mathbf{q}_2 \ \mathbf{q}_3)$ and find the coordinates of $\mathbf{x} = (3,0,0)^{\mathsf{T}}$ in this basis. Reconstruct $\mathbf{x}$ from those coordinates.

7. Let $\mathcal{S} = \operatorname{span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\} \subseteq \mathbb{R}^4$, where

$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ 2 \\ 2 \\ 1 \end{pmatrix}, \qquad \mathbf{a}_2 = \begin{pmatrix} 3 \\ 1 \\ 3 \\ -1 \end{pmatrix}, \qquad \mathbf{a}_3 = \begin{pmatrix} 6 \\ 2 \\ 4 \\ 2 \end{pmatrix}, \qquad \mathbf{y} = \begin{pmatrix} 4 \\ 3 \\ -2 \\ 4 \end{pmatrix}.$$

(a) Apply Gram–Schmidt, in the given order, to obtain an orthonormal basis of $\mathcal{S}$.

(b) Find the orthogonal projection $\mathbf{p}$ of $\mathbf{y}$ onto $\mathcal{S}$ and the distance from $\mathbf{y}$ to $\mathcal{S}$.

(c) Verify that the residual $\mathbf{y} - \mathbf{p}$ is orthogonal to each of $\mathbf{a}_1$, $\mathbf{a}_2$ and $\mathbf{a}_3$.

(d) A student computes $\sum_{i=1}^{3} (\mathbf{a}_i^{\mathsf{T}}\mathbf{y})\mathbf{a}_i$ as the projection. Explain why that formula does not apply to this basis.

8. A classifier has score $f(\mathbf{x}) = \mathbf{w}^{\mathsf{T}}\mathbf{x} + b$, where $\mathbf{w} = (2,-2,1)^{\mathsf{T}}$ and $b = -3$. Its decision boundary is $\mathcal{H} = \{\mathbf{x} : f(\mathbf{x}) = 0\}$.

(a) Find the distance from $\mathbf{p} = (5,-1,0)^{\mathsf{T}}$ to $\mathcal{H}$, and the closest point on $\mathcal{H}$.

(b) The labelled training points are $((3,0,0)^{\mathsf{T}}, +1)$, $((0,0,0)^{\mathsf{T}}, -1)$ and $((2,0,0)^{\mathsf{T}}, +1)$. Find their functional margins $m_i = y_i f(\mathbf{x}_i)$ and their smallest geometric margin $\min_i m_i / \|\mathbf{w}\|$.

(c) If $\mathbf{w}$ and b are both multiplied by 4, what happens to the boundary, the functional margins and the geometric margin?

9. Two candidate soft-margin SVMs have the following weight norms and functional margins on four labelled datapoints:

candidate	$\|\mathbf{w}\|^2$	m_1	m_2	m_3	m_4
P	2	2	1	0	-1
Q	10	1	1	1/2	1/2

(a) Find every hinge loss $\max\{0, 1 - m_i\}$ and its sum for each candidate.

(b) For the objective $J_C = \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_i \max\{0, 1 - m_i\}$ with $C > 0$, determine when each candidate has the smaller objective, including the tie case.

(c) Does a positive hinge loss necessarily mean a datapoint is misclassified? Explain using the table.

10. Which of the following matrices could be covariance matrices of a real dataset? Justify each answer. For each matrix that could be a covariance matrix, give a centered dataset in $\mathbb{R}^4$ with exactly that covariance matrix, using the $1/n$ convention. You may choose the number of datapoints.

$$\mathbf{C}_1 = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & -4 \end{pmatrix}, \quad \mathbf{C}_2 = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 3 \end{pmatrix}, \quad \mathbf{C}_3 = \begin{pmatrix} 1 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$\mathbf{C}_4 = \begin{pmatrix} 1 & -1 & 2 & 0 \\ -1 & 1 & -2 & 0 \\ 2 & -2 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{C}_5 = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 9 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

11. A dataset contains

$$\mathbf{x}_1 = \begin{pmatrix} 5 \\ 3 \\ 0 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} -1 \\ 3 \\ -2 \end{pmatrix}, \quad \mathbf{x}_4 = \begin{pmatrix} 1 \\ -1 \\ -4 \end{pmatrix}.$$

Apply PCA after centering, without rescaling features.

(a) Find the mean, centered data matrix and covariance matrix.

(b) Three eigenvectors of the covariance matrix are $(1, 0, 1)^\mathsf{T}$, $(0, 1, 0)^\mathsf{T}$ and $(1, 0, -1)^\mathsf{T}$. Find their eigenvalues, normalize them and order the principal directions.

(c) Using the first two principal directions from part (b), find the score vector and reconstruction of $\mathbf{x}_1$ in the original coordinates. State your direction choices.

(d) Find the fraction of total variance retained by two components and the smallest number of components that retains at least 90% of the variance.

(e) Find the average squared reconstruction error over all four datapoints when two components are retained. Distinguish this from the total squared reconstruction error.

12. A matrix $\mathbf{A} \in \mathbb{R}^{4\times 3}$ has the compact SVD $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathsf{T}}$, where

$$\mathbf{U} = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \mathbf{\Sigma} = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \mathbf{V}^{\mathsf{T}} = \begin{pmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \end{pmatrix}.$$

(a) Give the best approximations of rank at most 1 and rank at most 2 in Frobenius norm as explicit 4×3 matrices. Denote these approximations by $\mathbf{A}_1$ and $\mathbf{A}_2$, respectively.

(b) Find $\|\mathbf{A} - \mathbf{A}_1\|_F$, $\|\mathbf{A} - \mathbf{A}_2\|_F$ and their squares.

(c) What fraction of $\|\mathbf{A}\|_F^2$ is retained by $\mathbf{A}_2$, that is, what is $\|\mathbf{A}_2\|_F^2/\|\mathbf{A}\|_F^2$?

(d) Give the eigenvalues of $\mathbf{A}^{\mathsf{T}}\mathbf{A}$ and $\mathbf{A}\mathbf{A}^{\mathsf{T}}$, including multiplicities.

13. A recommender system predicts a rating table using $\widehat{\mathbf{R}} = \mathbf{P}\mathbf{Q}^{\mathsf{T}}$, with

$$\mathbf{P} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \mathbf{Q} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \\ 0 & 2 \\ 2 & -1 \end{pmatrix}.$$

(a) Compute $\widehat{\mathbf{R}}$. What is its shape and what is its rank?

(b) Only the ratings $(i, j, r_{ij}) = (1, 1, 3), (2, 3, 1), (3, 2, 5)$ are observed. Let $\Omega = \{(1, 1), (2, 3), (3, 2)\}$ be the set of observed index pairs. Compute the observed-entry squared loss $\sum_{(i,j)\in\Omega}(r_{ij} - \widehat{r}_{ij})^2$, where $\widehat{r}_{ij}$ is the (i, j) entry of $\widehat{\mathbf{R}}$.

(c) Explain why replacing every missing rating by zero and minimizing the squared error over the whole table changes the fitting problem.

(d) Using dense storage, if a table has m users and d items, how many scalar entries are stored in factors with k columns, compared with storing the full table? For $m = 1000$, $d = 500$ and $k = 10$, calculate both counts.

14. (*) Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{B} \in \mathbb{R}^{n\times p}$. Prove the inequality

$$\operatorname{rank}(\mathbf{AB}) \leq \operatorname{rank}(\mathbf{A}).$$

Use column spaces. Then give nonzero 2×2 matrices $\mathbf{A}, \mathbf{B}$ for which the product is the zero matrix, so that the inequality is strict.

15. (*) Let $\mathbf{A} \in \mathbb{R}^{m\times d}$.

(a) Prove that $\operatorname{Nul}(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = \operatorname{Nul}(\mathbf{A})$.

(b) Deduce that $\operatorname{rank}(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = \operatorname{rank}(\mathbf{A})$.

(c) Prove that $\mathbf{A}^{\mathsf{T}}\mathbf{A}$ is invertible if and only if the columns of $\mathbf{A}$ are linearly independent.